Abstract

The rapid growth of digital documents has made it increasingly difficult for organizations to efficiently access and utilize information stored across heterogeneous and multilingual document archives. Conventional keyword-based search systems often fail to capture semantic relationships between queries and document content, while general-purpose Large Language Models (LLMs) may generate fluent but unsupported responses when operating without reliable source grounding. These limitations become more significant in multilingual environments containing English and Indian-language documents, particularly when documents are distributed across native PDFs, DOCX files, and scanned documents requiring Optical Character Recognition (OCR). This project proposes a **Resource-Efficient, Citation-Grounded Multilingual Document Intelligence system based on Retrieval-Augmented Generation (RAG)** for private document collections.

The proposed system provides an end-to-end pipeline for document ingestion, OCR, language identification, page-aware text chunking, semantic indexing, retrieval, reranking, and evidence-grounded answer generation. Documents are processed while preserving metadata such as document identity, page number, language, and paragraph position, enabling responses to be accompanied by precise page-level source citations. The retrieval layer forms the core research component of the project through a controlled comparison of **direct multilingual embedding, translation-based retrieval, and hybrid dense-plus-lexical retrieval** strategies. Multilingual-E5 embeddings, PostgreSQL with pgvector, BM25-based lexical retrieval, and a multilingual cross-encoder reranker are incorporated to investigate how retrieval performance varies with language, query characteristics, document type, and OCR quality.

To improve reliability, the generation layer is restricted to retrieved evidence and incorporates an explicit **evidence-sufficiency and abstention mechanism**, allowing the system to decline answering when adequate supporting information is unavailable rather than generating an unsupported response. The system is designed for resource-constrained academic environments and can be delivered through a FastAPI backend and Next.js-based web interface.

The proposed evaluation framework measures **retrieval effectiveness, answer correctness, citation precision and recall, faithfulness, abstention precision, cross-lingual retrieval performance, OCR robustness, latency, and computational resource usage** across English, Hindi, Marathi, and other selected Indian languages. The project aims to establish an evidence-based understanding of which retrieval strategy provides the best balance of accuracy, trustworthiness, multilingual capability, and computational efficiency for private document intelligence. Its primary contribution is therefore not merely a multilingual chatbot, but a reproducible and deployable framework for **trustworthy, citation-grounded multilingual document retrieval and question answering**, addressing an identified gap in existing multilingual RAG research.